

The Search for Multiquark States at the Large Hadron Collider

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Abstract

Baryonic matter, of which we are primarily made of, is itself composed of three quarks. These baryons include the proton and neutron, which we know as the core constituents of the nucleus. There also exists the traditional mesons, which are composed of only two quarks. This begs the question: do particles composed of groups larger than a trio of quarks exist? The short answer is yes, with the first observation of a tetraquark - a particle composed of four quarks - being made in 2003 by the Belle Collaboration [4]. In this paper, the search for possible multiquark states using strange particles, namely, the K_s^0 meson and the Λ^0 baryon is done using data collected by the ATLAS detector of proton-proton collisions at a center-of-mass energy of $\sqrt{s} = 13.6$ TeV. Minimum bias data from 2015 having an integrated luminosity of 21.559 nb^{-1} is used. Secondary vertices are marked as either a K_s^0 or Λ^0 using strict cuts and their invariant mass and lifetimes are fitted for verification. A low energy resonant state of the $K_s^0 K_s^0$ invariant mass distribution was detected at 1524.40 ± 4.83 MeV which likely corresponds to the $f_2'(1525)$ state. With this known signal identified, a bump hunt is performed on the higher-energy range of the $K_s^0 K_s^0$, $K_s^0 \Lambda^0$ and $\Lambda^0 \Lambda^0$ invariant mass spectra in search for possible tetraquark, pentaquark and hexaquark states. The results of this bump hunt were inconclusive, with no signal exceeding a significance of 3σ . Further analysis is required to determine the presence of higher energy resonances.

1 Introduction

From the discovery of various exotic particles such as pions in the 50's to that of the Higgs boson in 2012, the advent of particle accelerators brought about new discoveries in particle physics, bolstering our grasp on the Standard Model - a gauge theory which describes three out of the four fundamental forces, that being the weak, strong, and electromagnetic forces [8], [1]. As the collision energies of particle accelerators steadily increase, new discoveries of higher mass particles can be made. Multiquarks are an example of these higher mass particles, being made up of collections of quarks whose number exceeds that of the usual two or three we observe in mesons and baryons respectively. Such particles are not forbidden by the Standard Model, and a proper identification of multiquark states may help in our understanding of quantum chromodynamics (QCD) - a core component of the standard model which describes the interactions between quarks and the elementary particles which mediates the strong interaction between them: gluons. Thus, this work searches for multiquark states through the possible decay products of tetraquarks, pentaquarks and hexaquarks. The decay channels are as follows:

$$\text{Tetraquark} : A \rightarrow K_s^0 K_s^0$$

$$\text{Pentaquark} : B \rightarrow K_s^0 \Lambda^0$$

$$\text{Hexaquark} : C \rightarrow \Lambda^0 \Lambda^0$$

Where the K_s^0 meson is a neutral kaon composed of a down quark (d) and an anti-strange quark ($\bar{s}$), and the Λ^0 , known as the Lambda baryon, is composed of an up, down and strange quark (uds). Quarks are known to possess one of the three fundamental color charges, of which there are three: Red, Green and Blue. According to the color confinement principle, a core tenet of QCD, color-charged particles such as quarks cannot be isolated, and thus quarks combine to form color-neutral hadrons called singlet states. Nothing forbids the creation of a singlet state composed of two quarks and two antiquarks ($qq\bar{q}\bar{q}$), a particle called a tetraquark which would be categorized as an exotic meson since, despite not being composed of a single quark-antiquark pair, it nonetheless possesses a meson's characteristic trait of being a hadron with a baryon number of zero. Similarly, nothing forbids the combination of a color-neutral baryon and a quark-antiquark pair ($qqqq\bar{q}$) which we would call a pentaquark. A combination of two baryons or three quark-antiquark pairs could also exist (q^6 or $q^3\bar{q}^3$), which are both examples of hexaquarks.

1.1 Event Reconstruction

The K_s^0 and Λ^0 are chosen as objects of study due to their decay products being easily identifiable within the ATLAS (A Torroidal LHC Apparatus) detector. The K_s^0 is known to decay into two charged pions ($\pi^+\pi^-$) with a branching ratio of $69.20 \pm 0.05\%$ [7] and the Λ^0 decays into a proton and a pion ($p^+\pi^-$) with a branching ratio of $63.9 \pm 0.5\%$ [6]. These charged particles are traced back to a secondary vertex (denoted $V0$) and the K_s^0 or Λ^0 invariant mass is reconstructed using the respective decay product's energies and momenta. The invariant mass of this two-particle decay (in natural units) is given by:

$$M = \sqrt{(E_1 + E_2)^2 - ||p_1 + p_2||^2}. \quad (1)$$

Once two secondary vertices are reconstructed within a single event, the process can be repeated to reconstruct the invariant mass of the primary vertex, which is the point of the $p - p$ collision, once again using equation 1. The resulting primary vertex invariant mass distribution can then be searched for possible multiquark resonances. Before this can be done, however, cuts must be placed on the reconstructed $V0$ s to reduce the background and ensure no misidentification of particles occur.

2 Methods

Each $V0$ is considered a potential K_s^0 and Λ^0 candidate and is subject to a series of cuts in order to properly categorize the secondary vertex. This is because the ATLAS detector obtains only data on the momentum and charge of the pions and protons. The trouble lies in differentiating the p^+ s from π^+ s, as the π^- s are

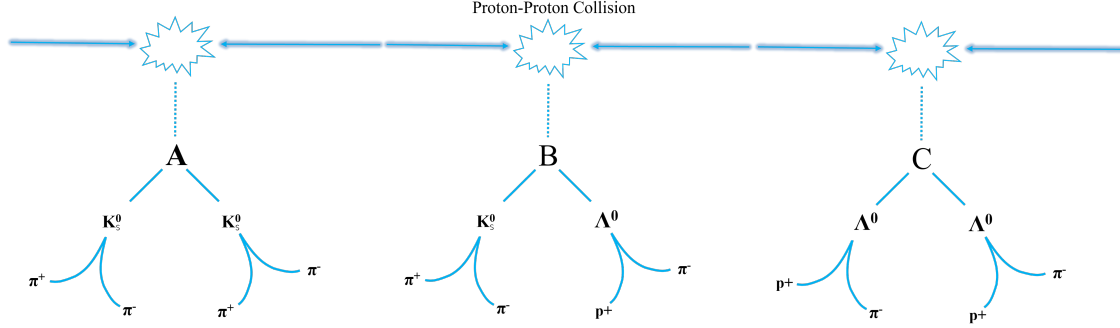
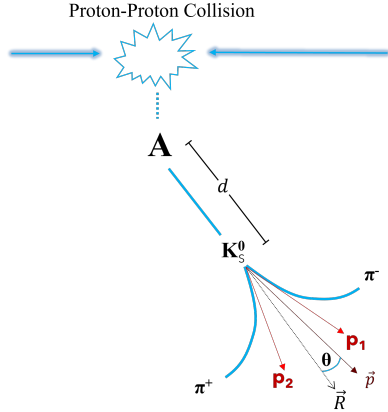


Figure 1: Tetraquark, pentaquark and hexaquark decay chains, respectively.

easily identifiable. In order to accomplish this, the invariant mass of each secondary vertex is computed twice: once with the proton mass, and another with the pion mass. The cuts are then used to determine which hypothesis was correct, as well as reduce the overall background. The cuts are as follows:



- Cut on pointing angle: $\cos(\theta)$
- Cut on transverse momentum: p_T
- Cut on distance travelled: d
- Cut on invariant mass: $m_{\pi^+\pi^-}$
- Cut on invariant mass: $m_{p^+\pi^-}$

With p_T and $\cos(\theta)$ being given by:

$$p_T = \sqrt{p_x^2 + p_y^2} \quad (2)$$

$$\cos(\theta) = \frac{\Delta x p_x + \Delta y p_y}{p_T d} \quad (3)$$

Figure 2: Kinematic variables of a $K_s^0 \rightarrow \pi^+\pi^-$ decay process.

Where Δx and Δy are the differences in x and y coordinates between the primary and secondary vertices and the equation for $\cos(\theta)$ comes from the dot product between the vectors $\vec{R}$ (the path vector whose magnitude is d) and $\vec{p}$, the reconstructed momentum, as per figure 2.

2.1 Cut on Pointing Angle

The most impactful cut which can be placed on a secondary vertex is the cut on the pointing angle. This cut removes a significant amount of the background as it is incredibly effective at ensuring the charged decay products originate from the same vertex. Requiring the cosine of the pointing angle to exceed 0.9998 removes a large fraction of the background while keeping most of the signal, as can be seen in figures 3 and 4 for the kaon and lambda respectively.

2.2 Cut on Transverse Momentum

The next cuts applied are those on the transverse momentum. Plotting p_T versus the K_s^0 and Λ^0 invariant mass distributions reveal clear signals around a mass of 497 MeV and 1115 MeV corresponding to their PDG mass values respectively. Figure 6 reveals a significant background below a transverse momentum of 500 MeV, whereas figure 5 shows the transverse momentum to be fairly localized near the kaon mass.

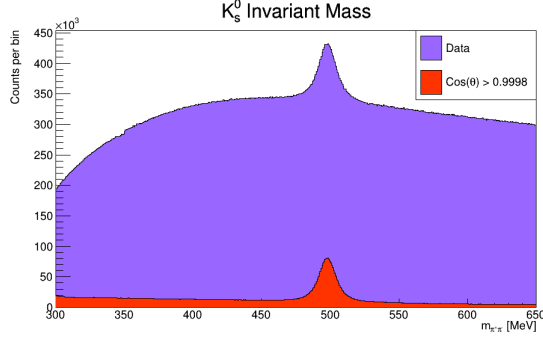


Figure 3: K_s^0 invariant mass histogram before and after cut on pointing angle.

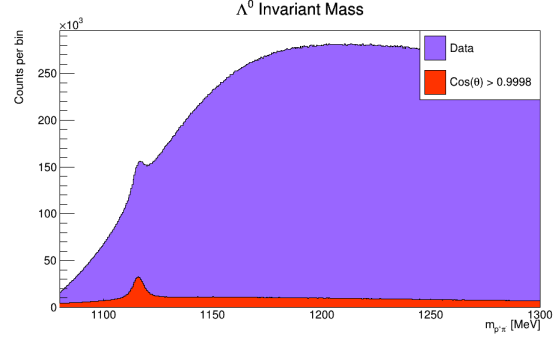


Figure 4: Λ^0 invariant mass histogram before and after cut on pointing angle.

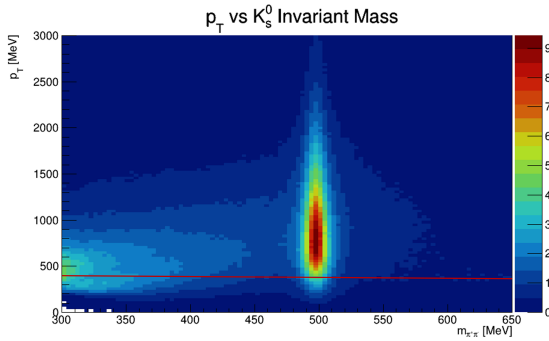


Figure 5: p_T versus K_s^0 invariant mass histogram with a cut for p_T values below 400 MeV.

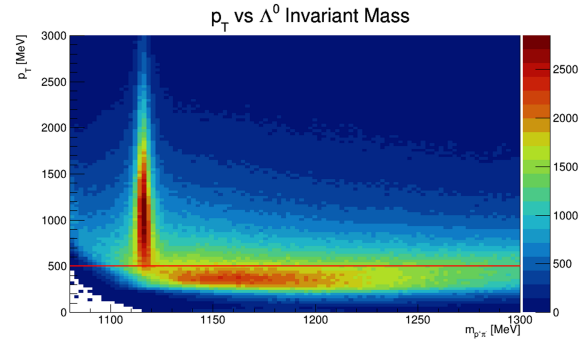


Figure 6: p_T versus Λ^0 invariant mass histogram with a cut for p_T values below 500 MeV.

2.3 Cut on Distance Travelled

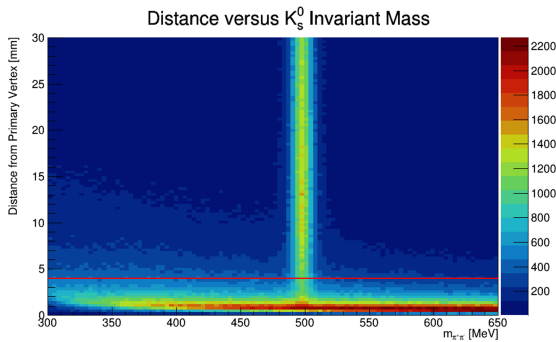


Figure 7: d versus K_s^0 invariant mass histogram with a cut for distances below 4 mm.

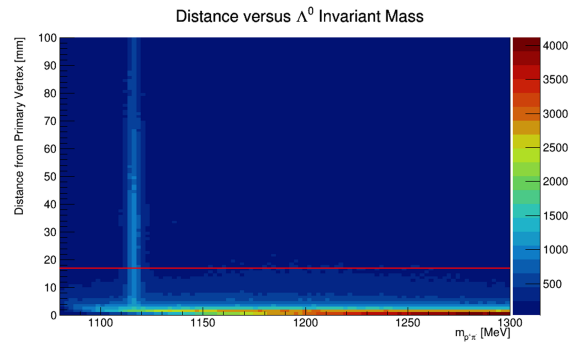


Figure 8: d versus Λ^0 invariant mass histogram with a cut for distances below 17 mm.

A cut on the distance travelled between the primary and secondary vertices is then placed on the data. This cut is effective at reducing background by removing particles whose lifetime is shorter than that of the K_s^0 or Λ^0 as the lifetime of a particle is related to its distance by

$$d = \beta\gamma c\tau, \quad (4)$$

where d is the average decay length, β is the ratio between the particle's velocity and the speed of light, $\gamma = 1/\sqrt{1 - \beta^2}$ is the Lorentz factor, and τ is the mean lifetime. Rearranging equation 4 and writing γ in terms of the momentum using $p = \gamma mv$ we obtain:

$$\tau = \frac{dm}{p} \quad (5)$$

Thus, as can be seen in figures 10 and 11, cuts were placed enforcing a minimum distance - and therefore lifetime on the secondary vertices to further reduce background.

A lifetime analysis is later performed, outlined in section 2.5, allowing for the placement of a cut on the maximum distance travelled by the secondary vertices.

2.4 Cut on K_s^0 and Λ^0 Mass

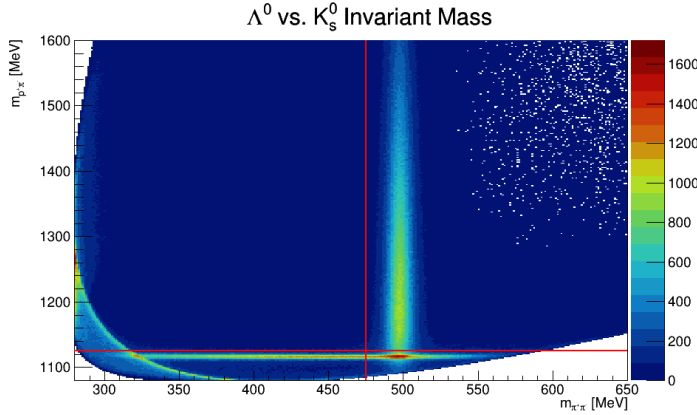


Figure 9: Λ^0 vs. K_s^0 invariant mass distributions with cuts to avoid misidentification of particles. K_s^0 candidates having a Λ^0 mass less than 1125 MeV and Λ^0 candidates having a K_s^0 mass greater than 475 MeV are rejected.

Another cut which can be placed is around the mass signal itself. This cut is likely the most obvious one to place, as the mass of the K_s^0 and Λ^0 are well known and a significant portion of the background seen in the plots of the previous sections can be removed by simply cutting around the mass signal. This is not without some difficulty, however, as certain secondary vertices happen to pass both the series of K_s^0 and Λ^0 cuts, including that of the mass signal cut. As can be seen on figure 9, there are secondary vertices whose reconstructed kaon mass $m_{\pi^+\pi^-}$ lies within the vertical streak near 497 MeV, but whose Lambda mass $m_{p+\pi^-}$ simultaneously lies within the horizontal streak near 1115 MeV. These secondary vertices can be mistaken as being both kaons and Lambdas and are

therefore rejected from this analysis due to their ambiguity.

2.5 Cut on K_s^0 and Λ^0 Lifetime

The lifetime distribution of all secondary vertices passing the previous cuts, as given by equation 5 can be fitted to a signal plus background model of the form:

$$y(t) = C_0 + C_b e^{-t/\tau_b} + C_s e^{-t/\tau_s} \quad (6)$$

Where C_0 and C_b are constants modelling the background having a lifetime of τ_b , while C_s and τ_s are the constant and lifetime of the signal respectively.

The point at which the signal contribution of the overall fit drops close to zero designates the ideal region to place a cut on the maximum lifetime of a secondary vertex, as it ensures maximal background reduction while minimizing signal reduction. This maximum lifetime is completely analogous to setting a maximum distance travelled. Furthermore, fitting this data according to equation 6 yields the particle's mean lifetime which can be compared to reference values to ascertain the reliability of this cut.

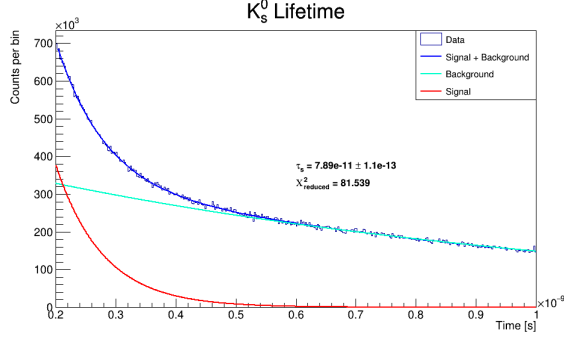


Figure 10: K_s^0 lifetime distribution with a fitted mean lifetime of 78.9 ± 0.1 ps compared to a PDG mean lifetime of 89.56 ± 0.03 ps [7].

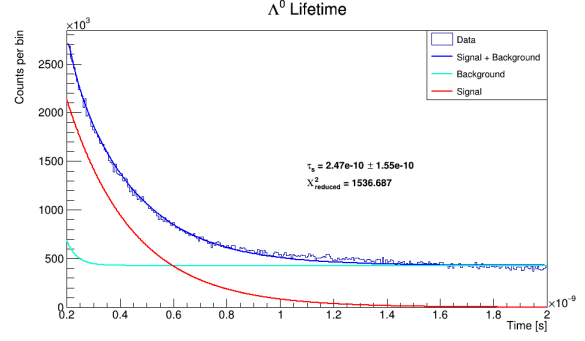


Figure 11: Λ^0 lifetime distribution with a fitted mean lifetime of 24.7 ± 20 ns compared to a PDG mean lifetime of 26.32 ± 0.02 ns [6].

2.6 Summary and Effect of All Cuts

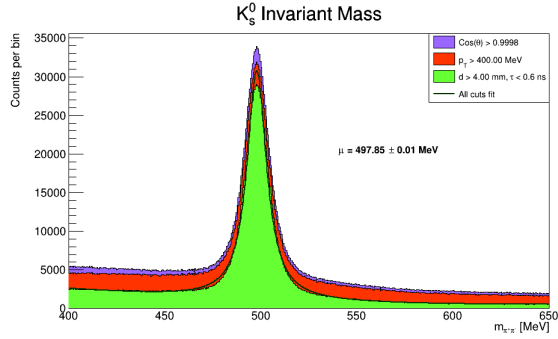


Figure 12: K_s^0 invariant mass histogram with all cuts progressively displayed. The distribution is fitted with a Breit-Wigner function for the signal and a quadratic function for the background. The signal is centered at 497.85 ± 0.01 MeV compared to the K_s^0 PDG mass of 497.61 ± 0.01 MeV [7]. Uncertainties on the obtained masses are statistical only.

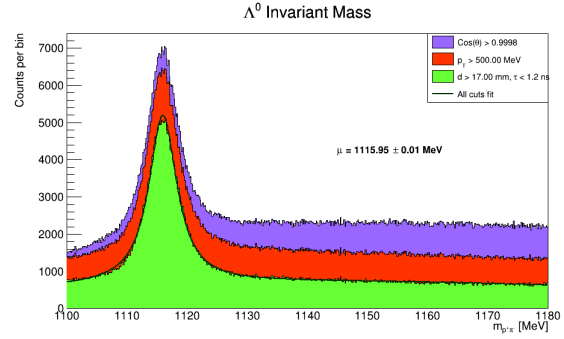


Figure 13: Λ^0 invariant mass histogram with all cuts progressively displayed. The distribution is fitted with a Breit-Wigner function for the signal and a quadratic function for the background. The signal is centered at 1115.95 ± 0.01 MeV compared to the Λ^0 PDG mass of 1115.683 ± 0.006 MeV [6]. Uncertainties on the obtained masses are statistical only.

With all these cuts put in place, the background is significantly reduced allowing for the reconstruction of multiquarks. Events are selected by requiring exactly two kaons, two Lambdas or one of each. The two secondary vertices are then used to compute the invariant mass of the multiquark, and the resulting distributions are plotted.

Table 1: Cuts Placed on Secondary Vertices

Variable	K_s^0 -min	K_s^0 -max	Λ^0 -min	Λ^0 -max
Pointing Angle - $\cos(\theta)$	0.9998	N/A	0.9998	N/A
Transverse Momentum - p_T [MeV]	400	N/A	500	N/A
Distance - d [mm]	4	N/A	17	N/A
K_s^0 Invariant Mass - $m_{\pi^+\pi^-}$ [MeV]	480	520	1125	N/A
Λ^0 Invariant Mass - $m_{p^+\pi^-}$ [MeV]	N/A	475	1105	1125
Lifetime - τ [ns]	N/A	0.6	N/A	1.2

3 Results

3.1 Tetraquark Invariant Mass Distribution

The tetraquark invariant mass distribution is searched for known resonances. These known resonances exist in the 1100-1800 MeV range, and have been found by the L3 experiment [2] and the HERA experiment [3] as can be seen in figures 15 and 16 respectively. The presence of these resonances in the reconstructed $K_s^0 K_s^0$ invariant mass distribution indicates an adequate reduction of the background, which potentially allows for the discovery of new resonances in the higher energy range. The $K_s^0 K_s^0$ invariant mass distribution of this analysis is shown in figure 14.

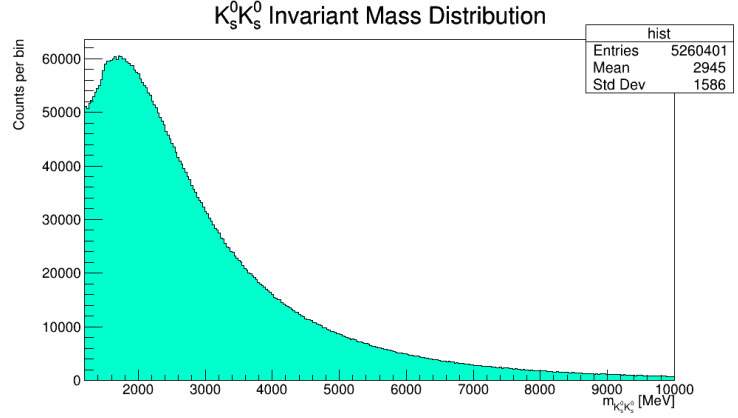


Figure 14: Tetraquark invariant mass distribution.

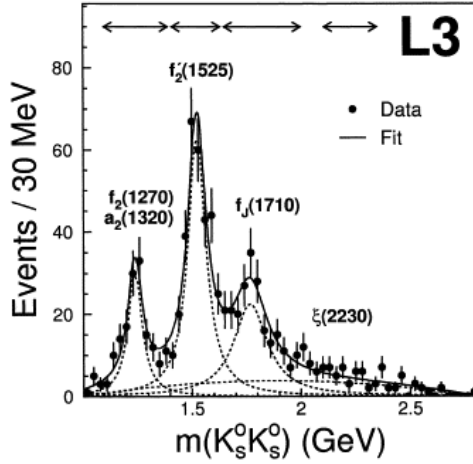


Figure 15: $K_s^0 K_s^0$ invariant mass distribution from the L3 experiment [2].

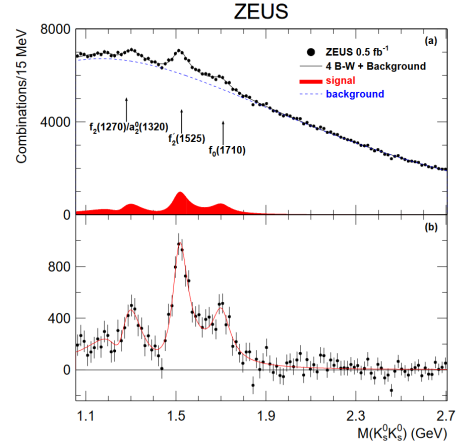


Figure 16: $K_s^0 K_s^0$ invariant mass distribution from the HERA experiment [3].

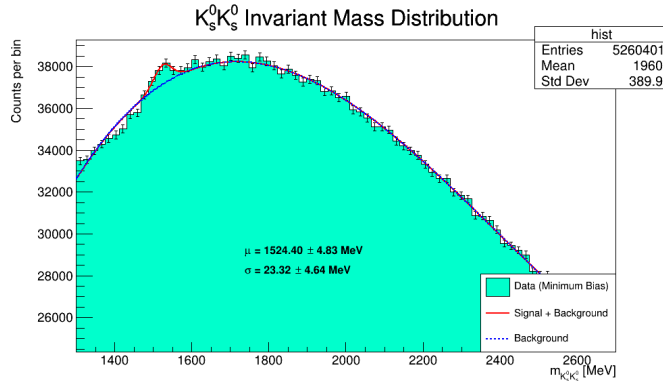


Figure 17: Tetraquark invariant mass distribution low energy resonance with a signal fitted with a Gaussian distribution having a mean of 1524.40 ± 4.83 MeV and a background fitted with a cubic polynomial.

A low energy resonance is detected at 1524.40 ± 4.83 MeV in the $K_s^0 K_s^0$ invariant mass distribution, as can be seen in figure 17. This likely corresponds to the $f_2'(1525)$ state, a well known resonance which has been observed by the L3 experiment [2] and the HERA experiment [3] as can be seen in figures 15 and 16. This resonance is the largest observed in the previously cited papers, and unfortunately the only one observed in this analysis. The search for higher energy resonances is ongoing, and efforts are being made to further reduce the background in hopes that other known resonances can be observed, indicating the possi-

bility of finding new resonances in the higher energy ranges.

3.2 Bump-hunt and significance

In searching for new resonances in the higher energy range, a bump hunt is performed on the $K_s^0 K_s^0$, $K_s^0 \Lambda^0$ and $\Lambda^0 \Lambda^0$ distributions. This is done by making multiple copies of the original distribution with a single bin missing, and fitting the background with an exponential function of polynomial order to the remaining data. The difference between the height of the missing bin and the evaluated background at that point is then taken to compute the significance of the bin in question. The significance is given by the distance over the error of the bin, which is simply given by the square root of the bin's value. This entire process is repeated for bin widths of 5, 50, 100, 200 and 500 MeV and the significance is plotted versus the bin center. This plot is then searched for bins having a significance greater than 3σ which persist across the various binnings. Such a scenario would indicate the presence of a potential discovery. The results of this bump hunt were inconclusive, with no bin having a significance exceeding that of 3σ , which is the lower limit for what may be considered a potential discovery in the field of particle physics. Further analysis is required to determine the presence of higher energy resonances.

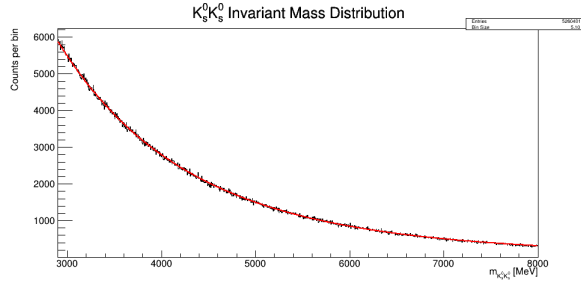


Figure 18: High energy range of $K_s^0 K_s^0$ invariant mass distribution with 5 MeV bin width fitted to an exponential background model.

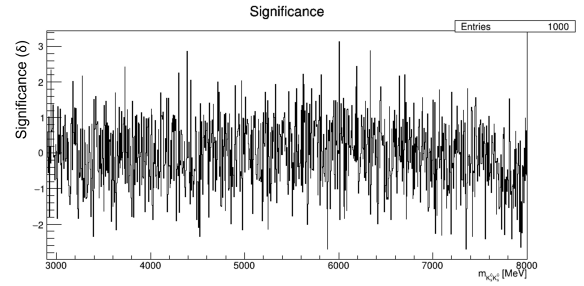


Figure 19: Significance of the $K_s^0 K_s^0$ invariant mass distribution with 5 MeV bin width.

3.3 Distribution of Significance

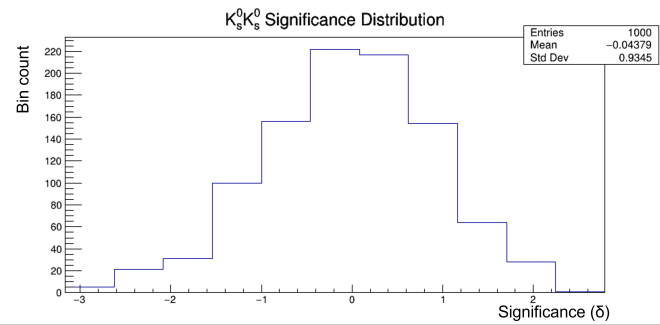


Figure 20: Significance distribution of the $K_s^0 K_s^0$ invariant mass distribution with 5 MeV bin width.

served in the higher energy range of the $K_s^0 K_s^0$ invariant mass distribution. The same can be said about the $K_s^0 \Lambda^0$ and $\Lambda^0 \Lambda^0$ invariant mass distributions.

The distribution of significance values for the $K_s^0 K_s^0$ invariant mass histograms are studied. This is effectively a projection of the significance values of the bump hunt onto the x-axis of new histogram. The distribution of significance values for the $K_s^0 K_s^0$ invariant mass with 5 MeV bin widths is shown in figure 20. This distribution is used to verify whether the significance is consistent across the various bin widths. A consistent normal distribution centered around zero is observed for each of the 5, 50, 100, 200 and 500 MeV bin widths, indicating that the significance is governed primarily by random fluctuations in the data. This is further evidence that no resonances were ob-

4 Conclusion

The search for multiquark states using minimum bias data from the ATLAS detector of proton-proton collisions at a center-of-mass energy of $\sqrt{s} = 13.6$ TeV has yielded a low energy resonance in the $K_s^0 K_s^0$ invariant mass distribution at 1524.40 ± 4.83 MeV, likely corresponding to the $f_2'(1525)$ state. This demonstrates an adequate reconstruction of the K_s^0 meson and Λ^0 baryon and reveals the effectiveness of the cuts placed on the secondary vertices to reduce background. The search for other known resonances is ongoing, with a bump hunt performed on the $K_s^0 K_s^0$, $K_s^0 \Lambda^0$ and $\Lambda^0 \Lambda^0$ invariant mass distributions. Further reduction of the background is required to ascertain the presence of higher energy resonances.

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